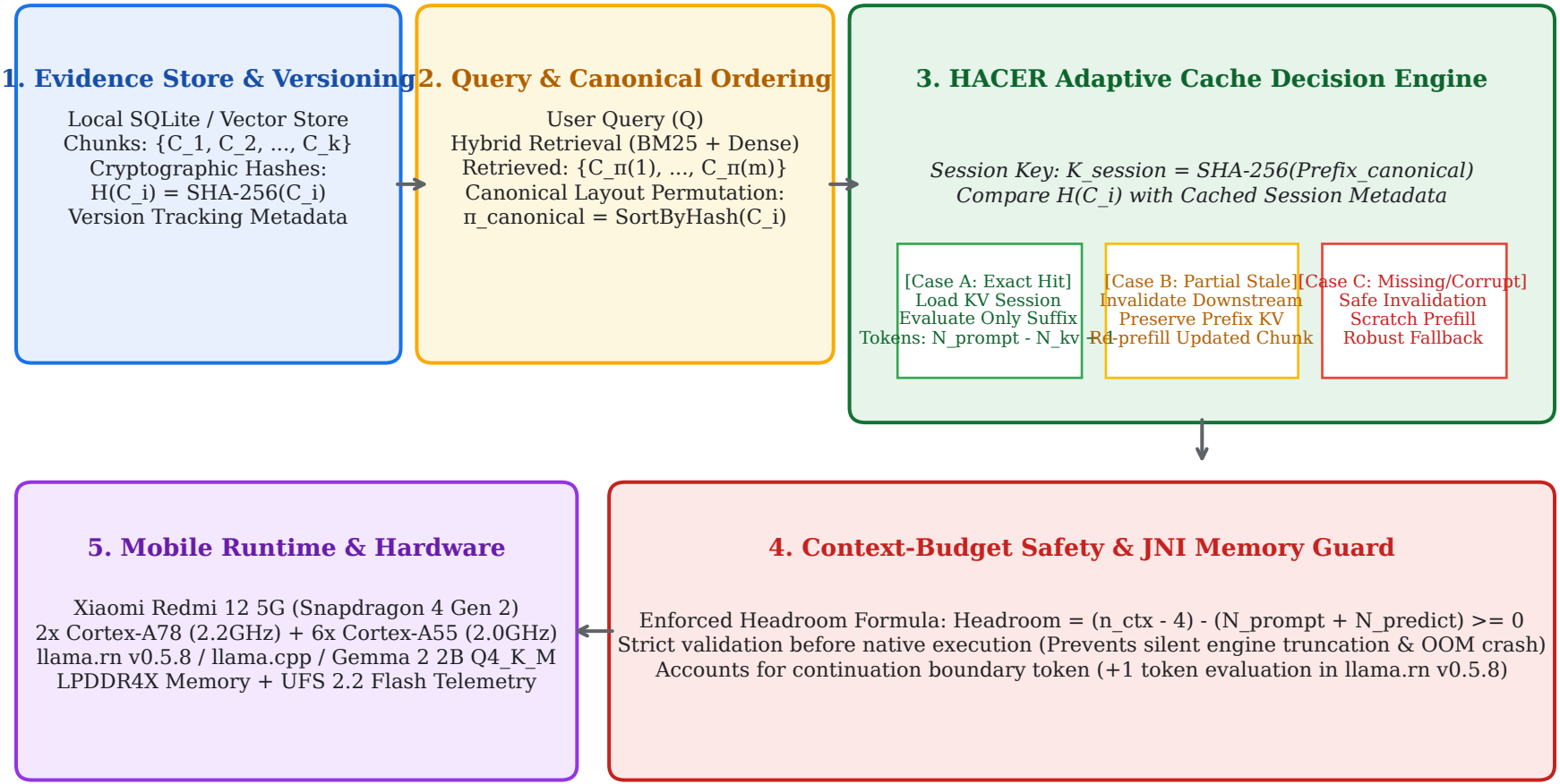
1. Evidence Store & Versioning
Local SQLite / Vector Store
Chunks: {C_1, C_2, ..., C_k}
Cryptographic Hashes:
H(C_i) = SHA-256(C_i)
Version Tracking Metadata
2. Query & Canonical Ordering
User Query (Q)
Hybrid Retrieval (BM25 + Dense)
Retrieved: {C_π(1), ..., C_π(m)}
Canonical Layout Permutation:
π_canonical = SortByHash(C_i)
3. HACER Adaptive Cache Decision Engine
Session Key: K_session = SHA-256(Prefix_canonical)
Compare H(C_i) with Cached Session Metadata
[Case A: Exact Hit]
Load KV Session
Evaluate Only Suffix
Tokens: N_prompt - N_kv
[Case B: Partial Stale]
Invalidate Downstream
Preserve Prefix KV
Re-prefill Updated Chunk
[Case C: Missing/Corrupt]
Safe Invalidation
Scratch Prefill
Robust Fallback
4. Context-Budget Safety & JNI Memory Guard
Enforced Headroom Formula: Headroom = (n_ctx - 4) - (N_prompt + N_predict) >= 0
Strict validation before native execution (Prevents silent engine truncation & OOM crash)
Accounts for continuation boundary token (+1 token evaluation in llama.rn v0.5.8)
5. Mobile Runtime & Hardware
Xiaomi Redmi 12 5G (Snapdragon 4 Gen 2)
2x Cortex-A78 (2.2GHz) + 6x Cortex-A55 (2.0GHz)
llama.rn v0.5.8 / llama.cpp / Gemma 2 2B Q4_K_M
LPDDR4X Memory + UFS 2.2 Flash Telemetry